

Quick Check

Lesson 4: Germination: Conditions Necessary and Types (Hypogeal vs Epigeal) in Maize H614 and Rosecoco Beans

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 4

Mr. Wafula's healthy, non-dormant maize and bean seeds were planted in the same field after the same rain, but some still failed to germinate. Which set lists the conditions a healthy, non-dormant seed needs in order to germinate?

A Water, oxygen and a suitable temperature

B Oxygen, soil and fertiliser

C Light, water and soil

D Water, light and a suitable temperature

Fatuma set up four jars, each with 20 Rosecoco bean seeds: water + air + 25 °C; dry; no air (boiled cooled water with an oil layer); and wet but kept at 4 °C. Which of these was kept the SAME in all four jars so that the test was fair?

- A The oxygen supply
- B The temperature
- C The water supply
- D The number of seeds (20) and the seed variety

A group is counting germinated Rosecoco bean seeds in a jar on Day 5. When should a seed be counted as GERMINATED?

- A** When the first leaves open
- B** When the seed coat becomes soft and swollen
- C** When the radicle has come out of the seed
- D** When the shoot appears above the soil

In Fatuma's set-up, the jar with water at 25 °C but no air (boiled cooled water with an oil layer) had only 1 of 20 seeds germinate. In the jar with water, air and 25 °C, 18 of 20 germinated. What does this comparison show?

- A** Water is enough on its own for germination
- B** Oxygen is needed, because the embryo uses it in aerobic respiration to release energy for growth
- C** Light is needed, because the oil layer blocked light
- D** Oxygen is not needed, because one seed still germinated

On Mr. Wafula's farm, 6 bean seeds and 3 maize seeds failed to germinate in a low patch that was waterlogged after 25 mm of rain. What is the best explanation?

- A** The rain washed away the light the seeds needed
- B** Water filled the soil pores, so the seeds got too little oxygen
- C** The soil became too cold for any seed to absorb water
- D** The seeds became dormant because they had too much water

Rosecoco beans show epigeal germination, and the cotyledons are carried above the soil. Which part of the seedling makes this happen?

- A** The scutellum grows upwards out of the soil
- B** The coleoptile pierces the soil and lifts the cotyledons
- C** The hypocotyl elongates as an arch and pulls the cotyledons above the soil
- D** The radicle grows upwards to push the cotyledons out

In the time-lapse results, the bean radicle came out on Day 3 and the bean shoot appeared above the soil on Day 8 (emergence). How many days passed between germination and emergence in the bean?

A 5 days

B 3 days

C 8 days

D 11 days

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).